



COMPUTATIONAL APPROACH TO ISO 7176-1 STATIC STABILITY VERIFICATION FOR WHEELCHAIR DESIGN

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UNDER THE GUIDANCE OF PROFESSOR GANESH BAPAT

What is ISO 7176-1

- This is a standard for static stability verification
- Testing of wheelchair's resistance to tipping under different conditions

Test conditions:

1. Tilt Orientation – Forward, Backward, Sideways (Left and Right)
2. Dummy – Included, Not included
3. Wheels – Locked, Not locked

This totals to 16 different conditions (Ex. Dummy Included, Locked and Forward)

Raising wheelchair quality in India using international testing standards

- 5 different wheelchairs (WC 1-5) were used for checking the ISO 7176 – 1 (Static Stability) and ISO 7176 - 8 (Fatigue Strength of Wheelchair) on a test set up
- For static stability the wheel chairs were tested for 8 different conditions, all including a 100kg dummy – forward, backward, left and right for both locked and unlocked scenarios
- For fatigue strength, only the wheelchairs which passed a double drum test (200,000 cycles) was put through for the drop test (6,666 cycles) – 5cm Drop

Table 2. ISO 7176-1 stability testing results.

		Tipping angles in degrees				
Stability direction		WC1	WC2	WC3	WC4	WC5
Forward	Downhill wheels unlocked	19°	20°	21°	20°	15°
	Downhill wheels locked	22°	11°	22°	17°	12°
Rearward	Rear wheels unlocked	18°	20°	Left wheel = 12°, right wheel = 18°	20°	24°
	Rear wheels locked	14°	10°	Left wheel = 9°, right wheel = 15°	14°	13°
Sideways	Left (unlocked)	22°	11°	9°	13°	15°
	Left (locked)	16°	15°	12°	8°	17°
Sideways	Right (unlocked)	22°	12°	6°	13°	4°
	Right (locked)	20°	9°	10°	10°	9°

Table 3. ISO 7176-8 durability testing results.

Wheelchairs	WC1	WC2	WC3	WC4	WC5
Double drum test	200,000	200,000	148,021	154,760	25,441
Curb drop test	6666	6666	–	–	–
Test results	Passed	Passed	Failed	Failed	Failed
Equivalent cycles	400,000	400,000	148,021	154,760	25,441
Cost in rupees	30,000	14,750	16,500	13,200	8700
Cost-effectiveness	13.33	27.12	8.97	11.72	2.92

Tipping Angle based on simple Geometry

- For forward and backward tipping angles of the wheel chair we only need to look at the wheelchair from one plane
- The parameters required – COG of Dummy, COG of Wheelchair, Wheelchair contact points with Floor
- Using simple trigonometry on the right you can see how to find the forward and backward tipping angles
- When Combined COG of wheelchair + dummy is beyond the contact points of the wheelchair, the wheelchair will tip.

Explanation

Finding tipping angle of wheelchair + Dummy
(Wheels Locked)

Finding combined C.G. of wheelchair & Dummy

point (0,0) is at the base of the rear wheel of the wheelchair. We are looking at backward tipping so one plane is enough, 3D image not required.

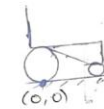
CG of wheelchair = (193.14, 391.23)

CG of Dummy = (243.14, 727)

all values in mm

mass of wheelchair = 19.75 kg = m_w

mass of dummy = 100 kg = m_d



$$\text{Combined CG} = \left(\frac{m_w x_w + m_d x_d}{m_w + m_d}, \frac{m_w y_w + m_d y_d}{m_w + m_d} \right)$$

$$= (235.72, 671.62)$$

To find tipping angle

(235.72, 671.62) → CG (combined)



$$\theta_f = \tan^{-1} \left(\frac{413.91 - 235.72}{671.62 - 0} \right) = 14.86^\circ$$

$$\theta_b = \tan^{-1} \left(\frac{235.72 - 0}{671.62 - 0} \right) = 19.34^\circ$$

Values from paper
12°

13°

Got some answers in my code

Python Code

The code uses the same principles from the previous slide

```
def combined_cg_mm(x1_mm, z1_mm, m1, x2_mm, z2_mm, m2):  
    """Mass-weighted combined CG (x_mm, z_mm)."""  
    mtot = m1 + m2  
    if mtot == 0:  
        raise ValueError("Total mass must be non-zero.")  
    x = (m1 * x1_mm + m2 * x2_mm) / mtot  
    z = (m1 * z1_mm + m2 * z2_mm) / mtot  
    print(f"x: {x}, z: {z}")  
    return x, z
```

```
def calculate_forward_tipping_angle(x_cg, z_cg, x_front_wheel, z_front_wheel):  
    a = math.sqrt((x_cg - x_front_wheel)**2 + (z_cg - z_front_wheel)**2)  
    theta = math.degrees(math.atan((z_cg - z_front_wheel) / (x_front_wheel - x_cg)))  
    tipping_angle = 0  
    while(x_cg < x_front_wheel):  
        theta += 0.1  
        tipping_angle += 0.1  
        x_cg = x_front_wheel - a * math.cos(math.radians(theta))  
    return tipping_angle
```

```
def calculate_backward_tipping_angle(x_cg, z_cg, x_back_wheel, z_back_wheel):  
    a = math.sqrt((x_cg - x_back_wheel)**2 + (z_cg - z_back_wheel)**2)  
    theta = math.degrees(math.atan((z_cg - z_back_wheel) / (x_cg - x_back_wheel)))  
    tipping_angle = 0  
    while(x_cg > x_back_wheel):  
        theta += 0.1  
        tipping_angle += 0.1  
        x_cg = x_back_wheel + a * math.cos(math.radians(theta))  
    return tipping_angle
```

Computational Approach using CAD Model and FEA Software

- The CAD Model was already created, 19.9 kgs close to the exact model which was 20 kgs
- COG from CAD Model (Wheelchair) – 198.14, 391.23, 302.81
- Using the CAD Model, simulations were done on ANSYS defining boundary conditions
- Different approaches were taken to understand the COG of Dummy – AI Model and Body Weight Distribution



FEA Boundary Conditions

- Fixed Support for Plank
- Angle of Standard Earth Gravity Changes to mimic plank incline in experimental setup
- Bonded Condition for all Joints
- Frictionless or Bonded based on locked or unlocked condition for contact region between axle and wheel
- Contact Patch between wheel and plank modelled to be rough surface to prevent sliding
- COG of Dummy where full mass concentration of 100kg is applied downwards in line with standard earth gravity

Dummy CAD Model

- Head and Neck – Volume: $7.465 \times 10^6 \text{ mm}^3$, Mass: 8.1 kg, Density: 1085.064 kg/m^3
- Thorax and Abdomen – Volume: $1.615 \times 10^7 \text{ mm}^3$, Mass: 45.5 kg, Density: 2817.337 kg/m^3
- Pelvis – Volume: $3.713 \times 10^6 \text{ mm}^3$, Mass: 14.2 kg, Density: 3824.401 kg/m^3
- Thighs – Volume: $7.939 \times 10^6 \text{ mm}^3$, Mass: 20 kg, Density: 2519.209 kg/m^3
- Legs – Volume: $1.143 \times 10^7 \text{ mm}^3$, Mass: 9.3 kg, Density: 813.648 kg/m^3
- Feet – Volume: $2.709 \times 10^6 \text{ mm}^3$, Mass: 2.9 kg, Density: 1070.506 kg/m^3
- Total Mass = 100kg
- COG – (192.18, 733, 302.81)
- Reference – Biomechanics and Motor Control of Human Movement by David A. Winter

Based on AI Model & Body Weight Distribution Model

- The COG of the Dummy was taken to be 235mm from the seat upwards, 240mm from the backrest and at the centre of the wheelchair from the other axis due to symmetry based on AI Model
- Experimental Results (Wheels Locked with Dummy): Forward – 12 degrees
Backward – 13 degrees
Left – 17 degrees
Right – 9 degrees
- Simulation Results: Forward – 13 degrees
Backward – 15 degrees
Sideward (Left and Right) – 18 degrees
- The COG of the Dummy was taken to be 241mm from the seat upwards, 189.04mm from the backrest and at the centre of the wheelchair from the other axis due to symmetry based on body weight distribution model
- Checked for Forward only and tipped after more than 15 degrees, showing more error than AI Model

Future Work

- Experimental Testing without dummy could be done as the experimental results from the paper, “Raising wheelchair quality in India using international testing standards”, do not match with the results that we got using the different methods of computational approach
- If we have values of the experimental tests without the dummy, we can use those values as the reference for the computational approach
- Perfect the simulation with or without dummy and complete the 16 conditions with WC -5
- Check if the Geometric Method or figure out if there are any mistakes in the Geometric Method
- Move towards ISO 7176 – 8 Fatigue Testing and figure out a computational approach towards testing this as well
- Finally, extend the study to further wheel chairs

THANK YOU